argmin and monotonicity

CSCI 270

model space

$$y = \theta \cdot x + \epsilon$$

loss function

$$L(\theta) = (.72\theta_S + .06\theta_L - 20)^2 + (.34\theta_S + .25\theta_L - 28)^2 + (.17\theta_S + .57\theta_L - 41)^2$$

training data

	x_S saturation	x_L lightness	y age
	.72	.06	20
	.34	.25	28
	.17	.57	41

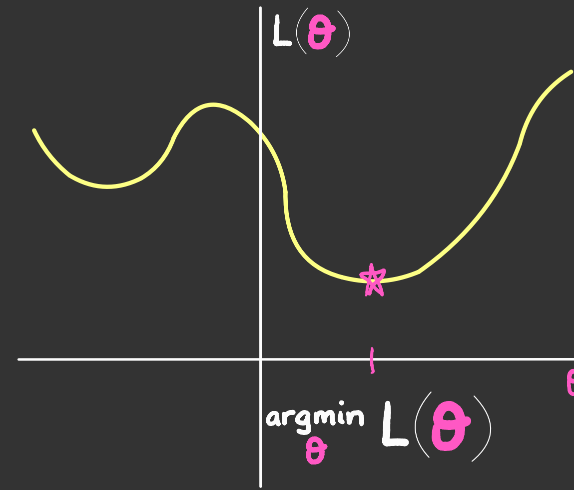
optimization

$$\underset{\theta}{\operatorname{argmin}}\ L(\theta)$$

let's discuss

argmin

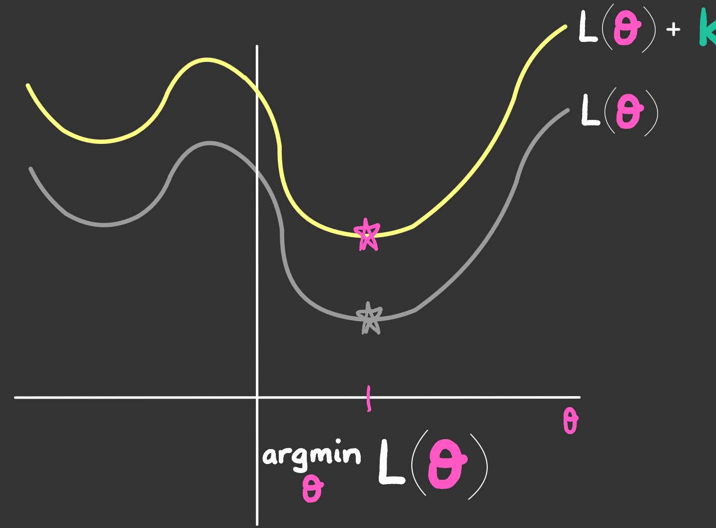
if this is $L(\theta)$:



what is:

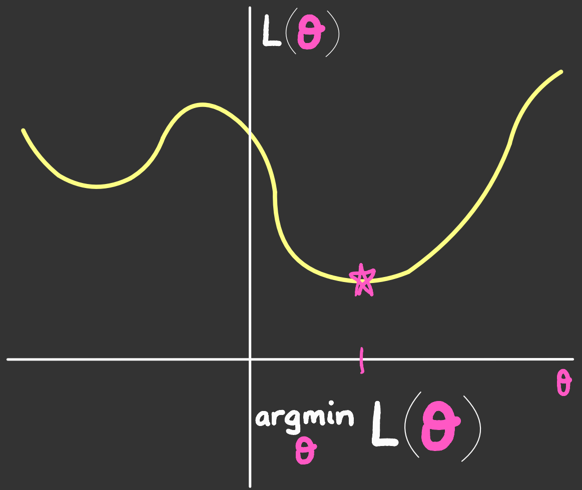
$$\underset{\theta}{\operatorname{argmin}}\left[L(\theta) + k\right]?$$

if this is $L(\theta)$:



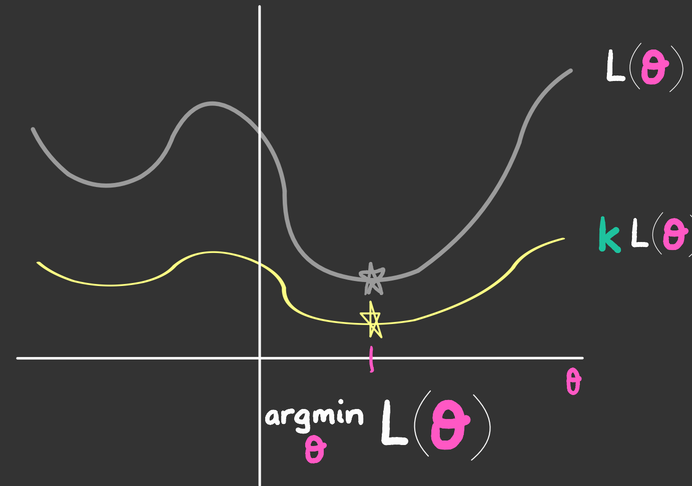
$$\underset{\theta}{\operatorname{argmin}} \left[L(\theta) + k \right] = \underset{\theta}{\operatorname{argmin}} L(\theta)$$

if this is $L(\theta)$:



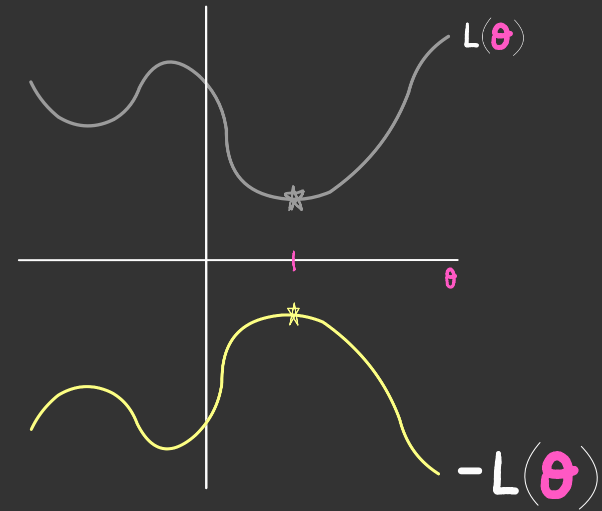
what is: $\underset{\theta}{\operatorname{argmin}}\, k L(\theta)$?

if this is $L(\theta)$:



$$\underset{\theta}{\operatorname{argmin}}\, k L(\theta) = \begin{cases} \underset{\theta}{\operatorname{argmin}}\, L(\theta) & \text{if } k > 0 \\ & \end{cases}$$

if this is $L(\theta)$:



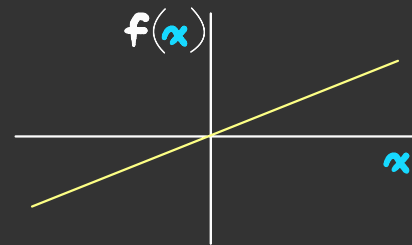
$$\underset{\theta}{\operatorname{argmin}}\, k\, L(\theta) = \begin{cases} \underset{\theta}{\operatorname{argmin}}\, L(\theta) & \text{if } k > 0 \\ \underset{\theta}{\operatorname{argmax}}\, L(\theta) & \text{if } k < 0 \end{cases}$$

this phenomenon can be generalized

let $f(x) = kx$

$k\,L(\theta)$

positive

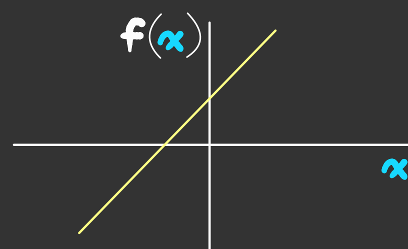


then

$f(L(\theta)) = k\,L(\theta)$

let $f(x) = x + k$

$L(\theta) + k$



then

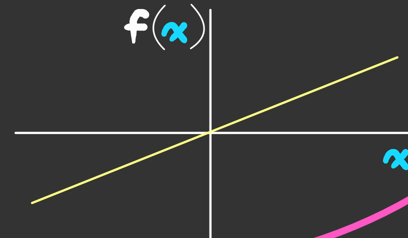
$f(L(\theta)) = L(\theta) + k$

both are monotonically increasing functions

$$kL(\theta)$$

positive

let $f(x) = kx$

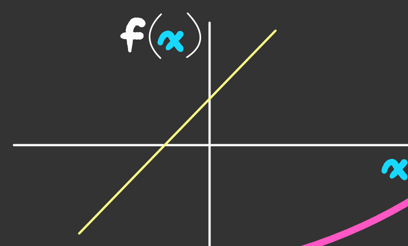


then

$$f(L(\theta)) = kL(\theta)$$

$$L(\theta) + k$$

let $f(x) = x + k$



then

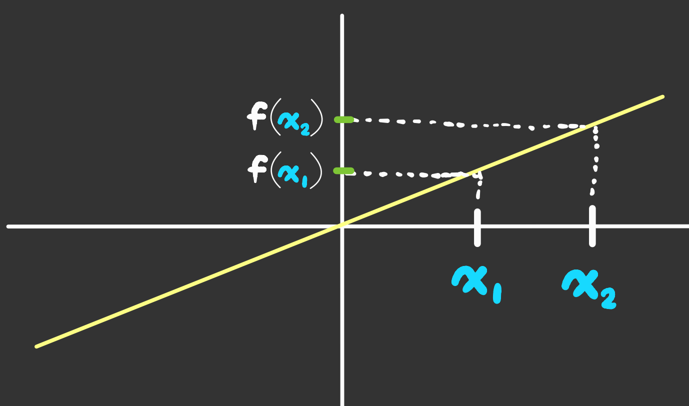
$$f(L(\theta)) = L(\theta) + k$$

monotonically increasing

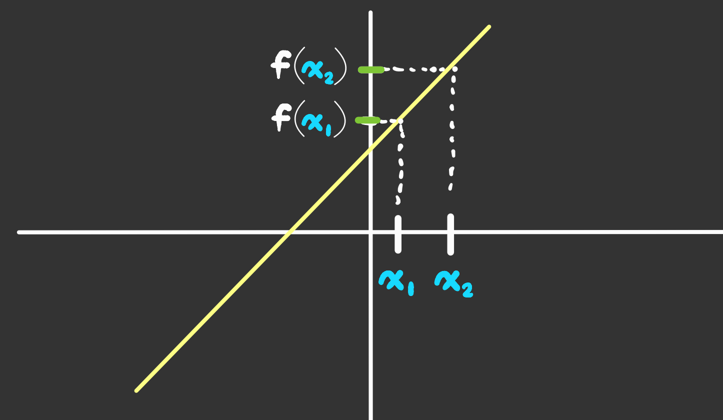
if $x_1 < x_2$, then $f(x_1) < f(x_2)$

$f(x) = kx$

positive

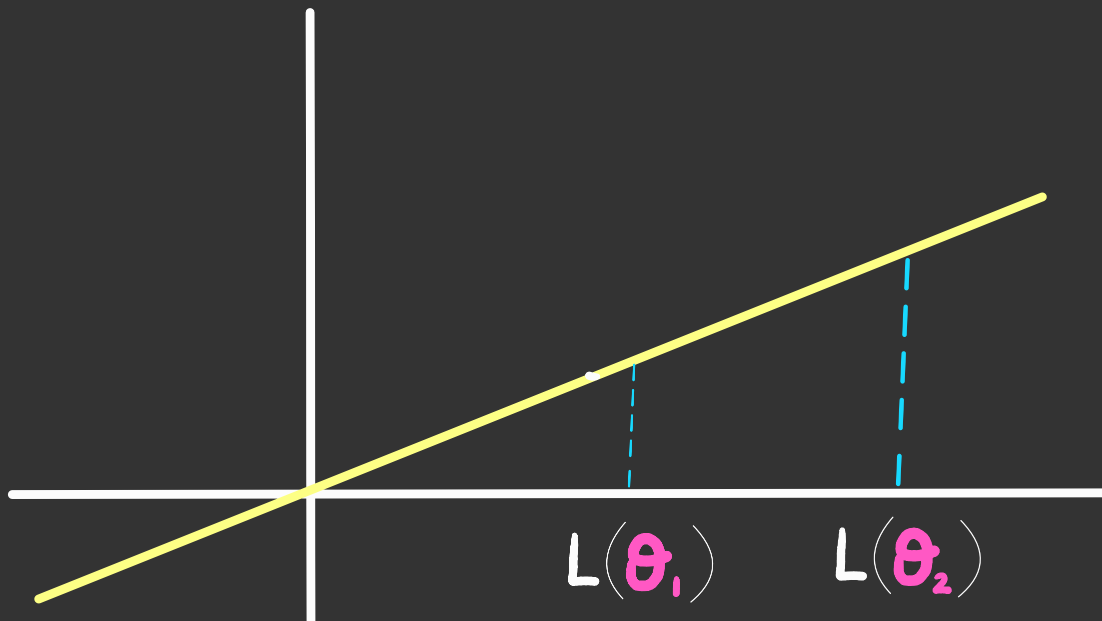


$f(x) = x + k$

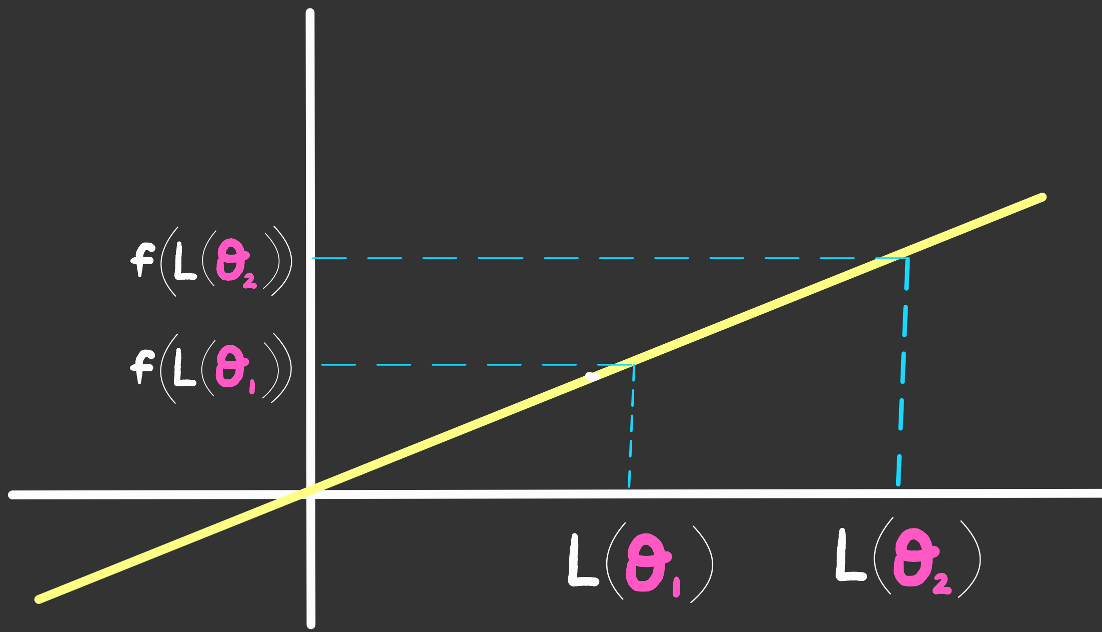


$$\text{if } \theta_1 = \underset{\theta}{\operatorname{argmin}}\, L(\theta), \text{ then } L(\theta_1) < L(\theta_2)$$

for any other θ_2

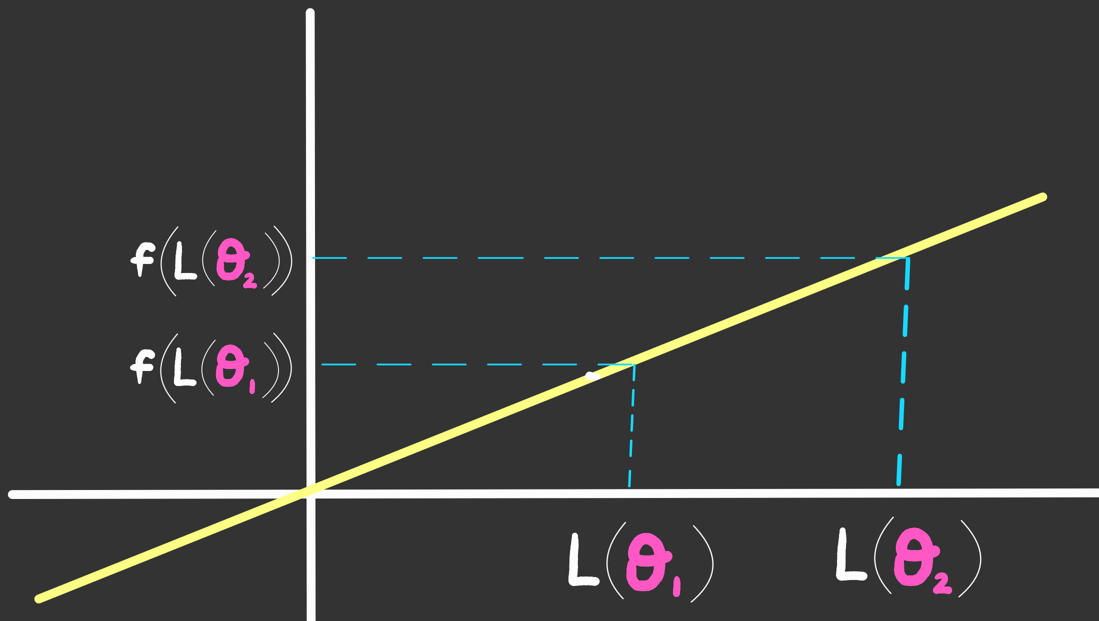


so $f(L(\theta_1)) < f(L(\theta_2))$ for any other θ_2

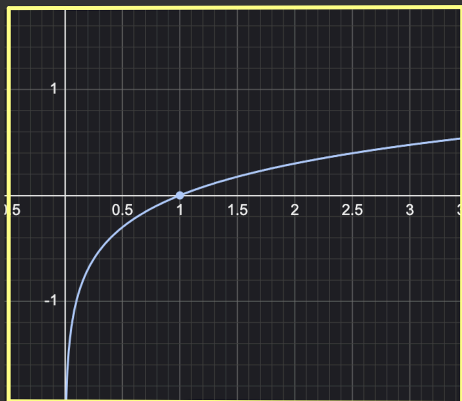


$$\text{so } \underset{\theta}{\operatorname{argmin}} f(L(\theta)) = \underset{\theta}{\operatorname{argmin}} L(\theta)$$

when f is monotonically increasing

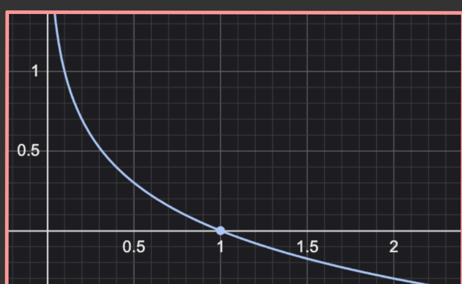


other similar facts



when f is monotonically increasing

- $\underset{\theta}{\operatorname{argmin}}\, f(L(\theta)) = \underset{\theta}{\operatorname{argmin}}\, L(\theta)$
- $\underset{\theta}{\operatorname{argmax}}\, f(L(\theta)) = \underset{\theta}{\operatorname{argmax}}\, L(\theta)$



when f is monotonically decreasing

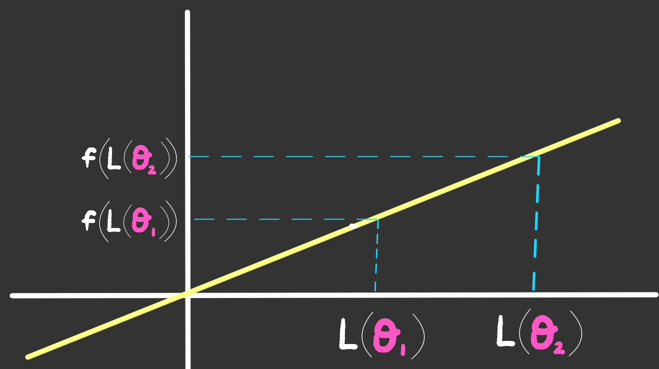
- $\underset{\theta}{\operatorname{argmin}}\, f(L(\theta)) = \underset{\theta}{\operatorname{argmax}}\, L(\theta)$
- $\underset{\theta}{\operatorname{argmax}}\, f(L(\theta)) = \underset{\theta}{\operatorname{argmin}}\, L(\theta)$

(provided these exist)

why should we care that

$$\underset{\theta}{\operatorname{argmin}}\, f\big(L(\theta)\big) = \underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

when f is **monotonically increasing**



?

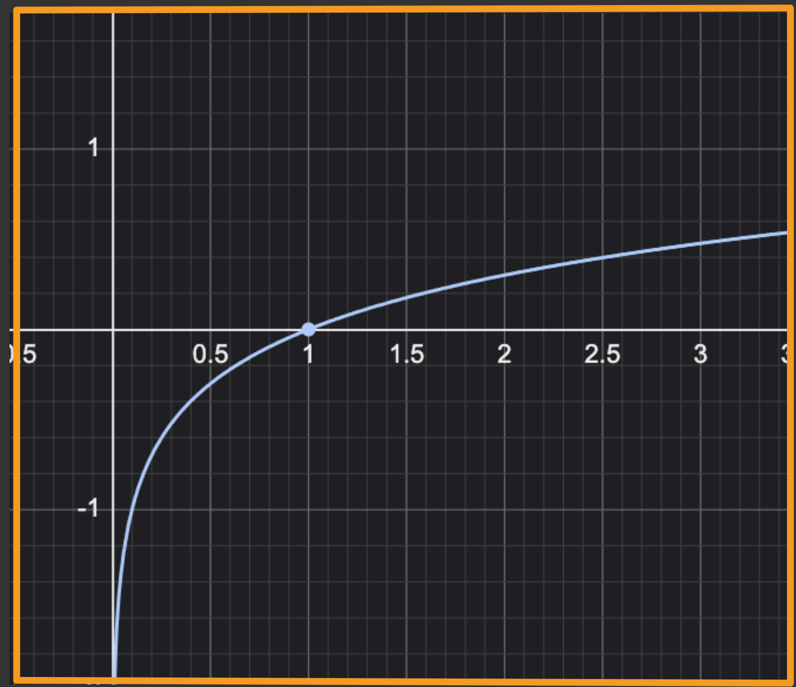
why should we care that

$$\operatorname*{argmin}_{\theta} \boxed{f(L(\theta))} = \operatorname*{argmin}_{\theta} \boxed{L(\theta)} \text{ ?}$$

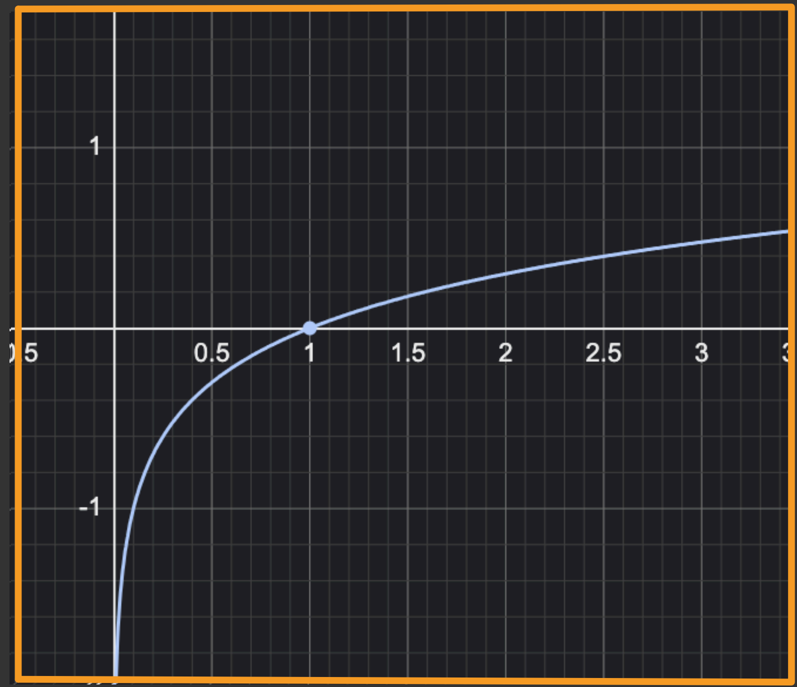
when f is monotonically increasing

sometimes this is easier to optimize

log is the superstar of monotonic functions



log is monotonically increasing over positive numbers



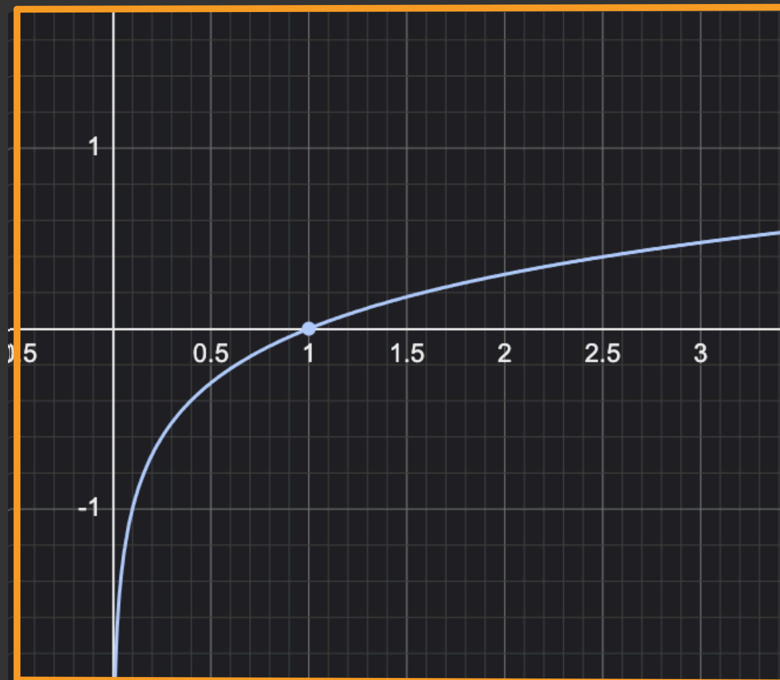
when f is monotonically increasing

$$\underset{\theta}{\operatorname{argmin}}\, f(L(\theta)) = \underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

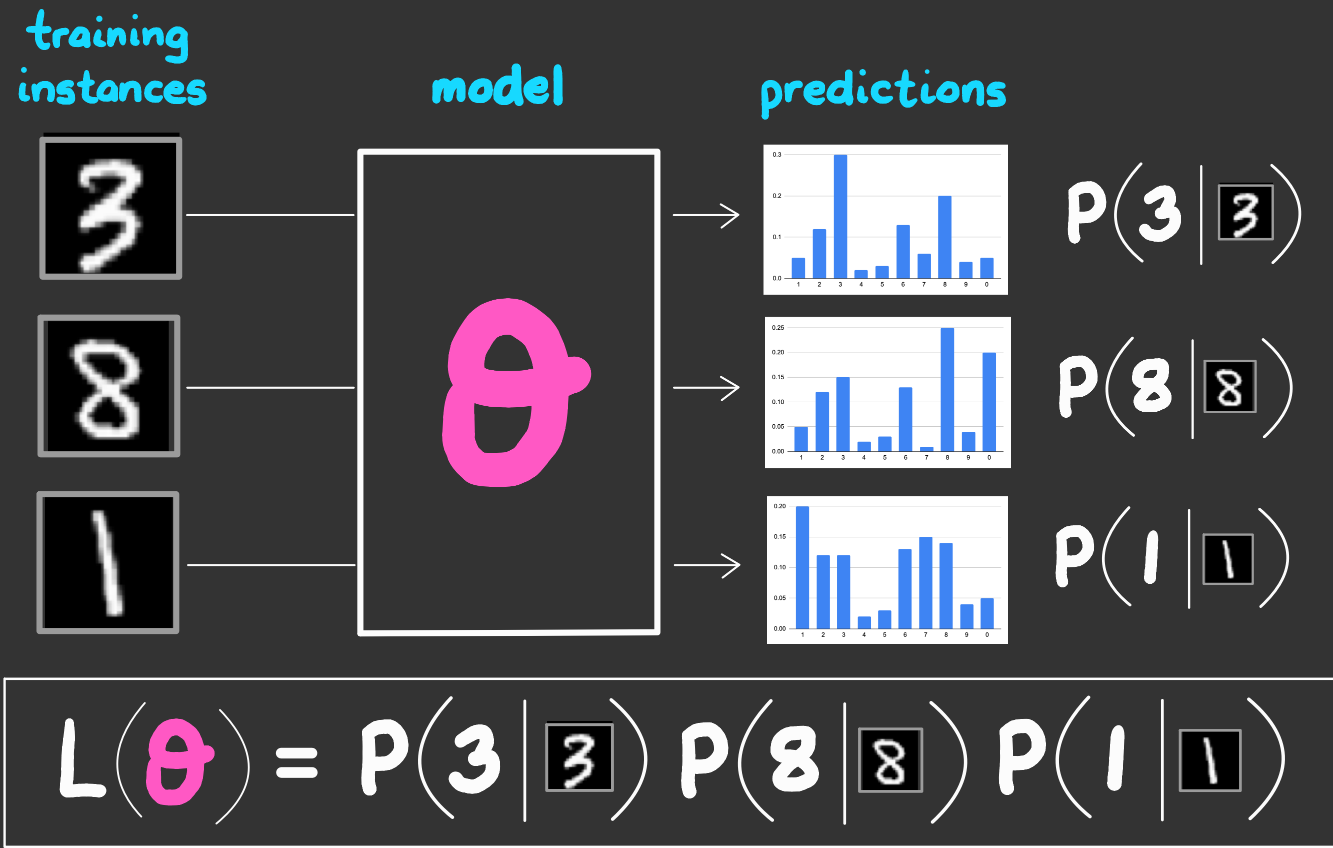
therefore:

$$\underset{\theta}{\operatorname{argmin}}\, \log(L(\theta)) = \underset{\theta}{\operatorname{argmin}}\, L(\theta)$$

when the range of L is positive



log is particularly useful with probabilities



why is this a bad loss function?

$L(\theta) = P_\theta(3 \mid$  $)\,P_\theta(8 \mid$  $)\,P_\theta(1 \mid$  $)$

why is this a bad loss function?

$L(\theta) = P_\theta(3 \mid$  $)\,P_\theta(8 \mid$  $)\,P_\theta(1 \mid$  $)$

```python
x = 1.0
for i in range(1100):
    x = x * 0.5
    print(x)
```

underflow

```
0.5
0.25
0.125
0.0625
0.03125
.
.
.
8e-323
4e-323
2e-323
1e-323
5e-324
0.0
0.0
0.0
0.0
0.0
0.0
```

luckily: $\underset{\theta}{\mathrm{argmax}} \log(L(\theta)) = \underset{\theta}{\mathrm{argmax}}\, L(\theta)$

$L(\theta) = P_\theta(3 \mid$  $)\, P_\theta(8 \mid$  $)\, P_\theta(1 \mid$  $)$

$\log L(\theta) = \log \big[P_\theta(3 \mid$  $)\, P_\theta(8 \mid$  $)\, P_\theta(1 \mid$  $) \big]$

$= \log P_\theta(3 \mid$  $) + \log P_\theta(8 \mid$  $) + \log P_\theta(1 \mid$  $)$

two equivalent loss functions

$$L(\theta) = P_\theta(3 \mid \boxed{3})\,P_\theta(8 \mid \boxed{8})\,P_\theta(1 \mid \boxed{1})$$

$$L(\theta) = \log P_\theta(3 \mid \boxed{3}) + \log P_\theta(8 \mid \boxed{8}) + \log P_\theta(1 \mid \boxed{1})$$

```python
x = 1.0
for i in range(1100):
    x = x * 0.5
    print(x)
```

underflow

```
0.5
0.25
0.125
0.0625
0.03125
.
.
.
8e-323
4e-323
2e-323
1e-323
5e-324
0.0
0.0
0.0
0.0
0.0
0.0
```

```python
from math import log
x = 0.0
for i in range(7):
    x = x + log(0.5)
    print(x)
```

totally ok

```
-1.3862943611198906
-2.0794415416798357
-2.772588722239781
-3.4657359027997265
-4.1588830833596715
-4.852030263919617
.
.
.
-756.2235739908822
-756.9167211714421
-757.609868352002
-758.3030155325619
-758.9961627131217
-759.6893098936816
-760.3824570742415
-761.0756042548014
-761.7687514353613
-762.4618986159212
```